Class-4

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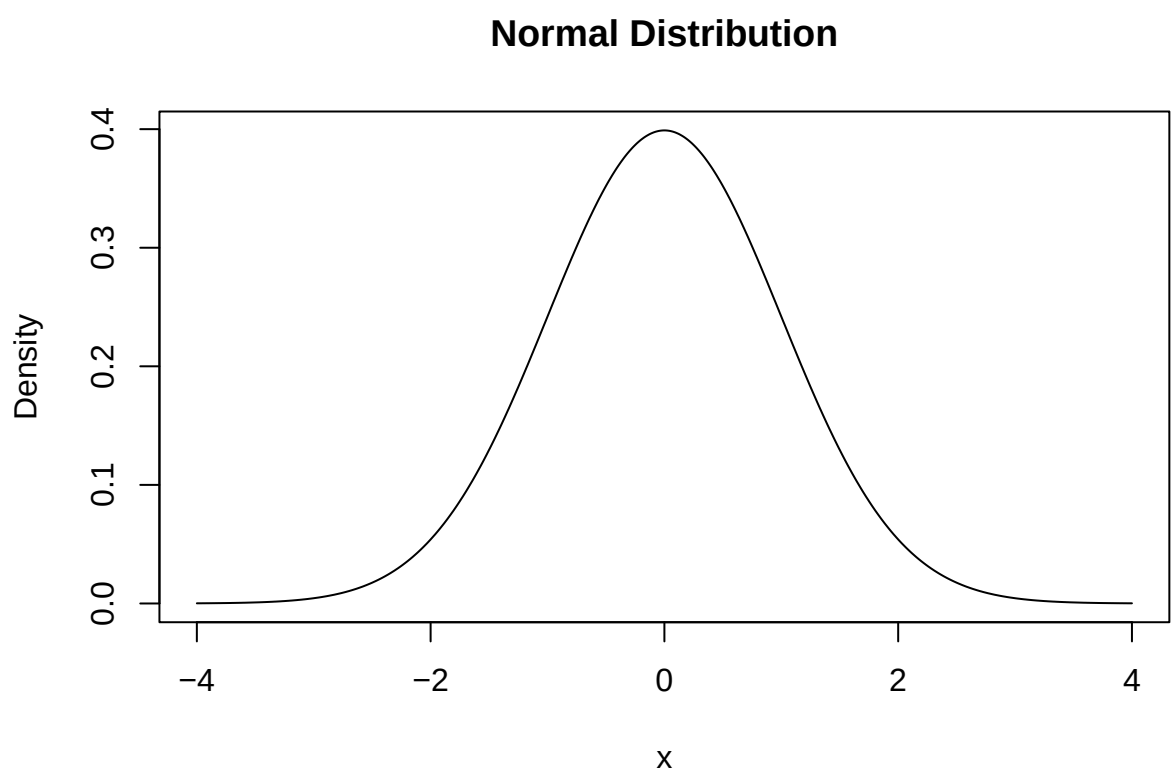
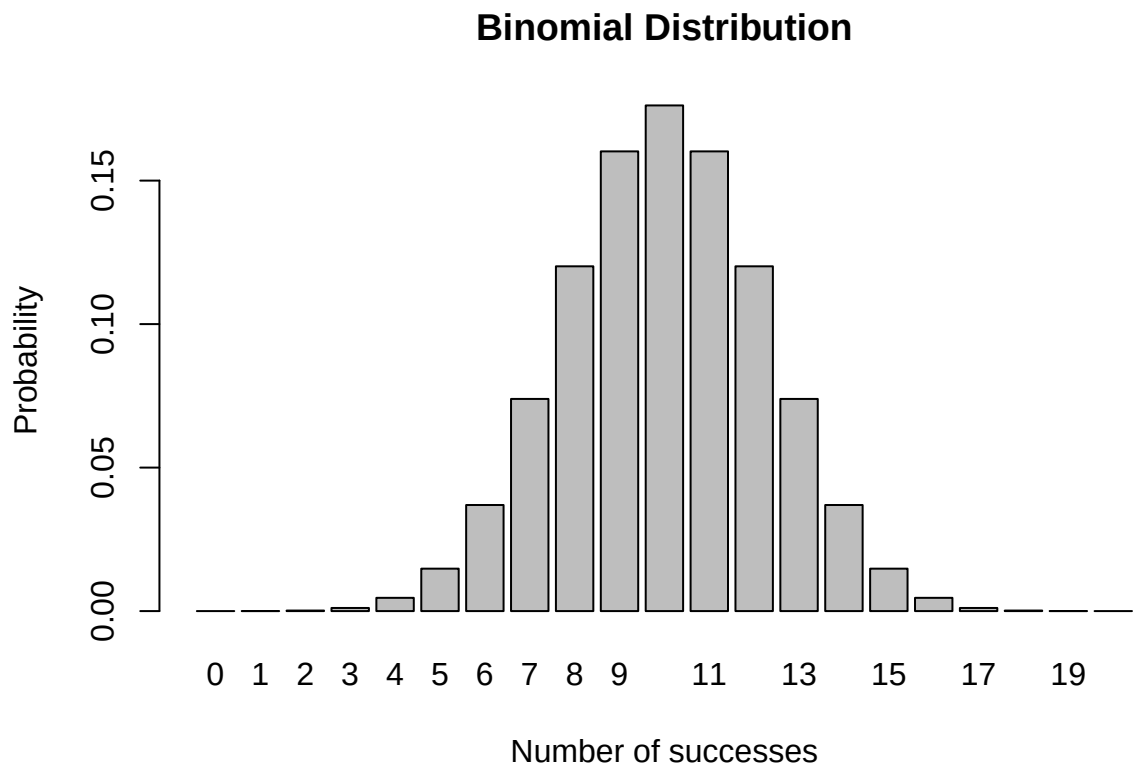
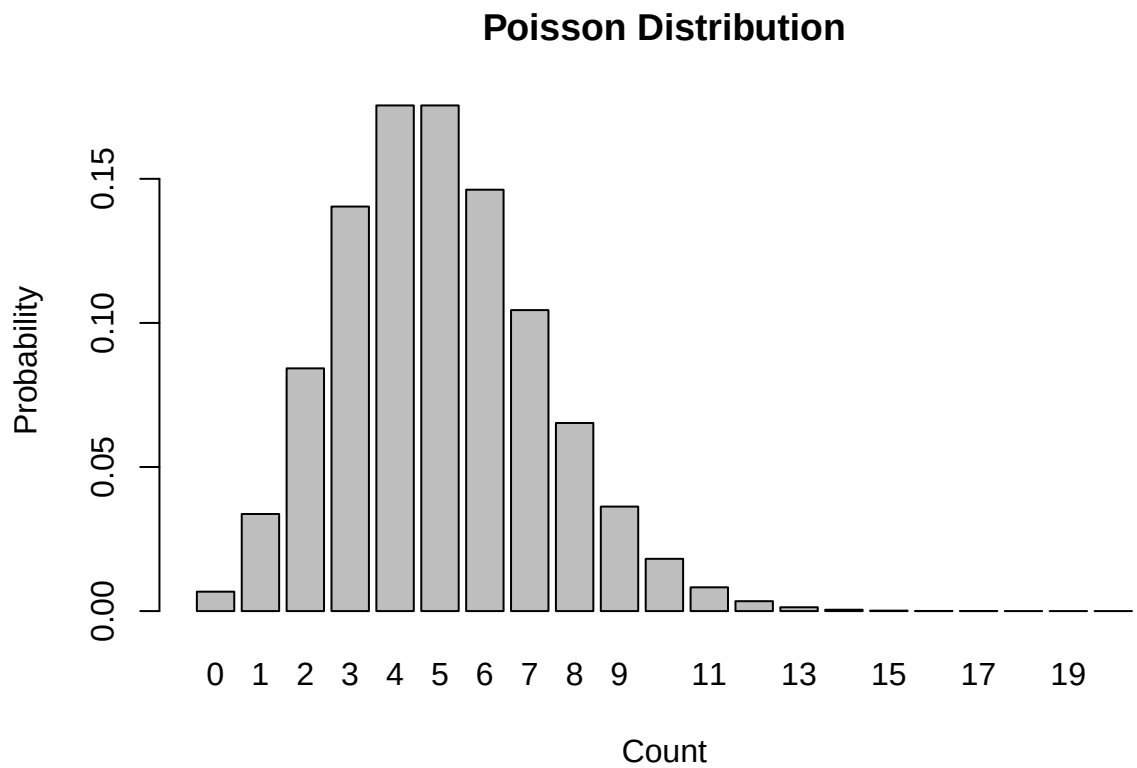
```r
x <- seq(-4, 4, length = 1000)
y <- dnorm(x, mean = 0, sd = 1)

plot(x, y, type = "l",
     main = "Normal Distribution",
     ylab = "Density", xlab = "x")
```




```r
x <- 0:20
y <- dbinom(x, size = 20, prob = 0.5)

barplot(y, names.arg = x,
        main = "Binomial Distribution",
        xlab = "Number of successes", ylab = "Probability")
```




```r
x <- 0:20
y <- dpois(x, lambda = 5)

barplot(y, names.arg = x,
        main = "Poisson Distribution",
        xlab = "Count", ylab = "Probability")
```




```r
set.seed(1)

normal_data  <- rnorm(10000, mean = 0, sd = 1)
binom_data   <- rbinom(10000, size = 20, prob = 0.5)
pois_data    <- rpois(10000, lambda = 5)
```

```r
summary(normal_data)
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## -3.671300 -0.673394 -0.015929 -0.006537  0.677660  3.810277
```

```r
sd(normal_data)
```

```
## [1] 1.012356
```

```r
summary(binom_data)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.000   8.000  10.000   9.974  11.000  18.000
```

```r
sd(binom_data)
```

```
## [1] 2.21784
```

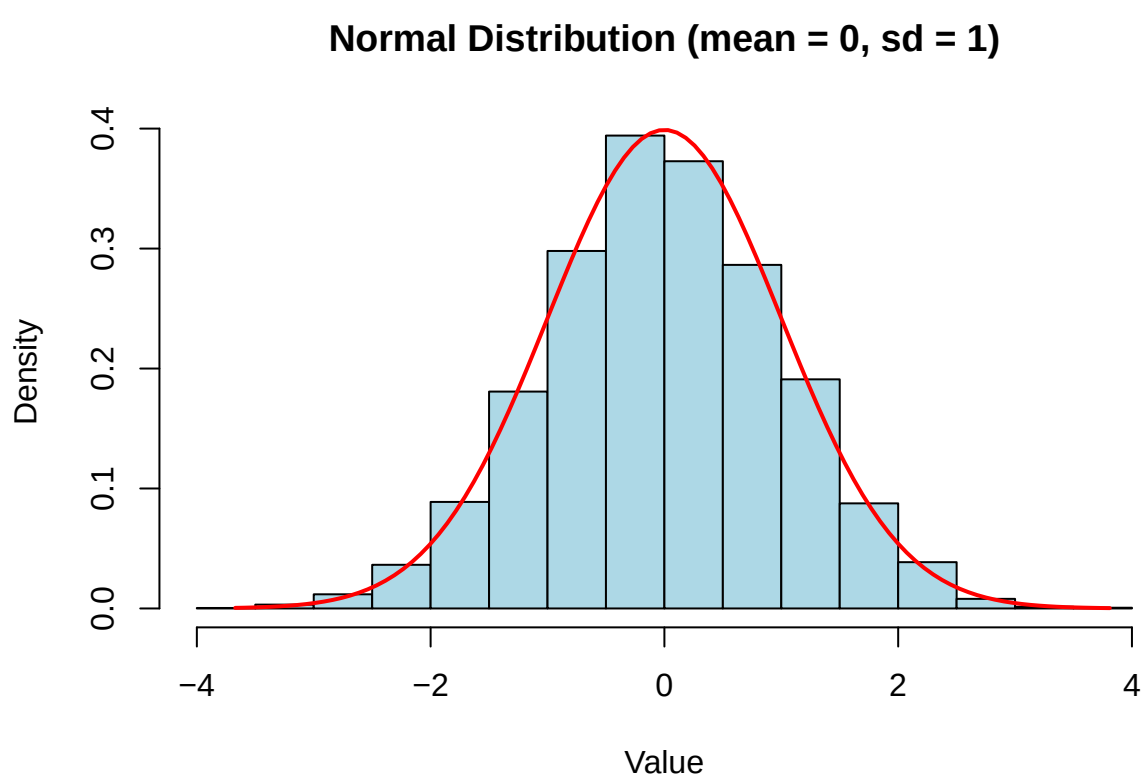
```r
summary(pois_data)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   3.000   5.000   5.007   6.000  15.000
```

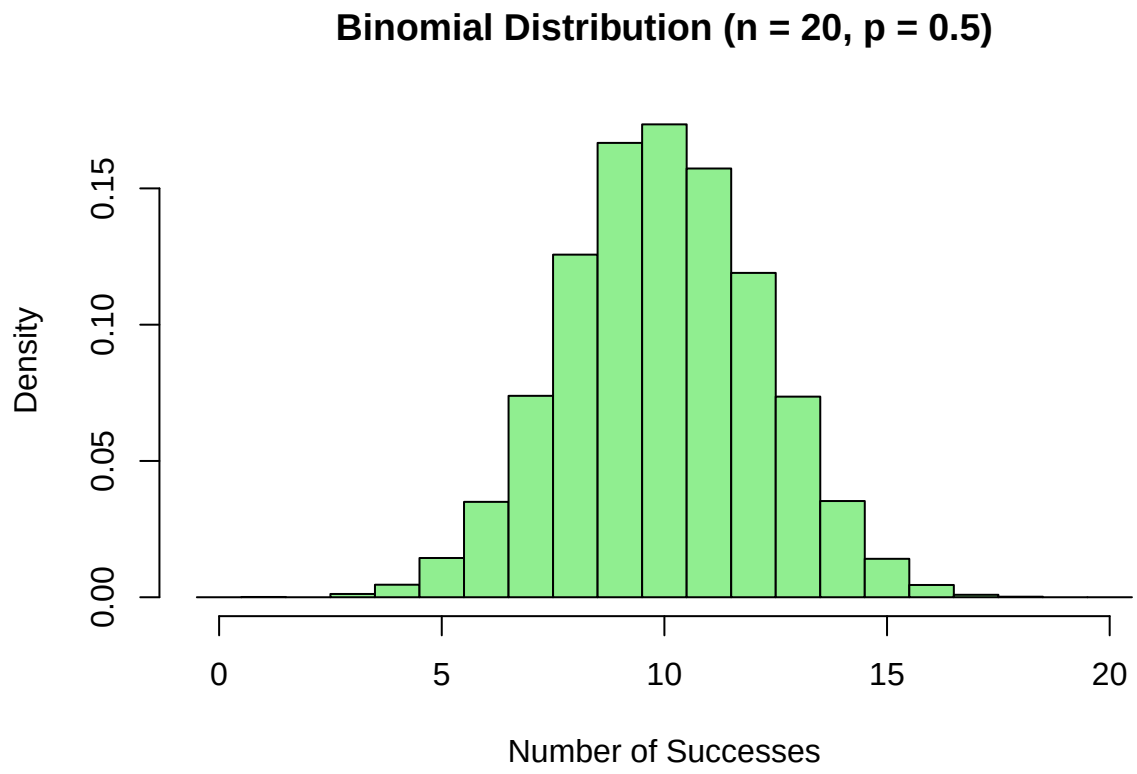
```r
sd(pois_data)
```

```
## [1] 2.235117
```

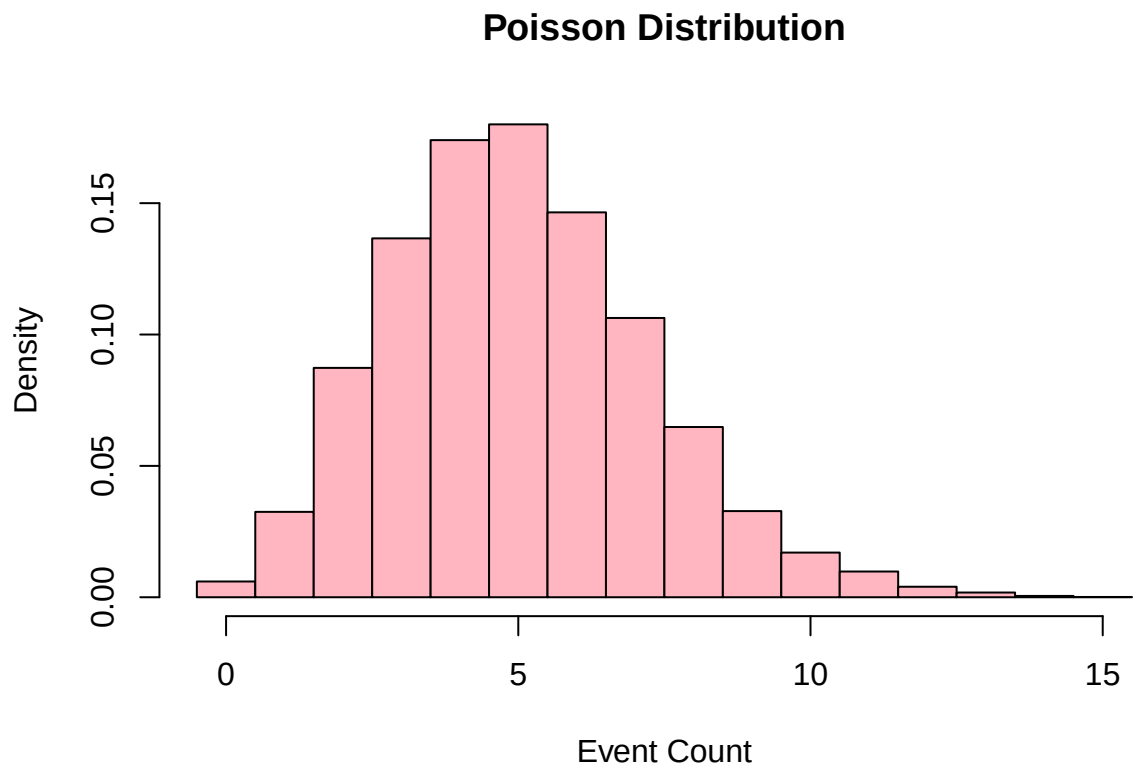
```r
hist(normal_data,
     probability = TRUE,
     col = "lightblue",
     main = "Normal Distribution (mean = 0, sd = 1)",
     xlab = "Value")

# Overlay theoretical curve
x <- seq(min(normal_data), max(normal_data), length = 100)
lines(x, dnorm(x, mean = 0, sd = 1), col = "red", lwd = 2)
```




```r
hist(binom_data,
     probability = TRUE,
     col = "lightgreen",
     main = "Binomial Distribution (n = 20, p = 0.5)",
     xlab = "Number of Successes",
     breaks = seq(-0.5, 20.5, by = 1))
```




```r
hist(pois_data,
     probability = TRUE,
     col = "lightpink",
     main = "Poisson Distribution",
     xlab = "Event Count",
     breaks = seq(-0.5, max(pois_data) + 0.5, by = 1))
```

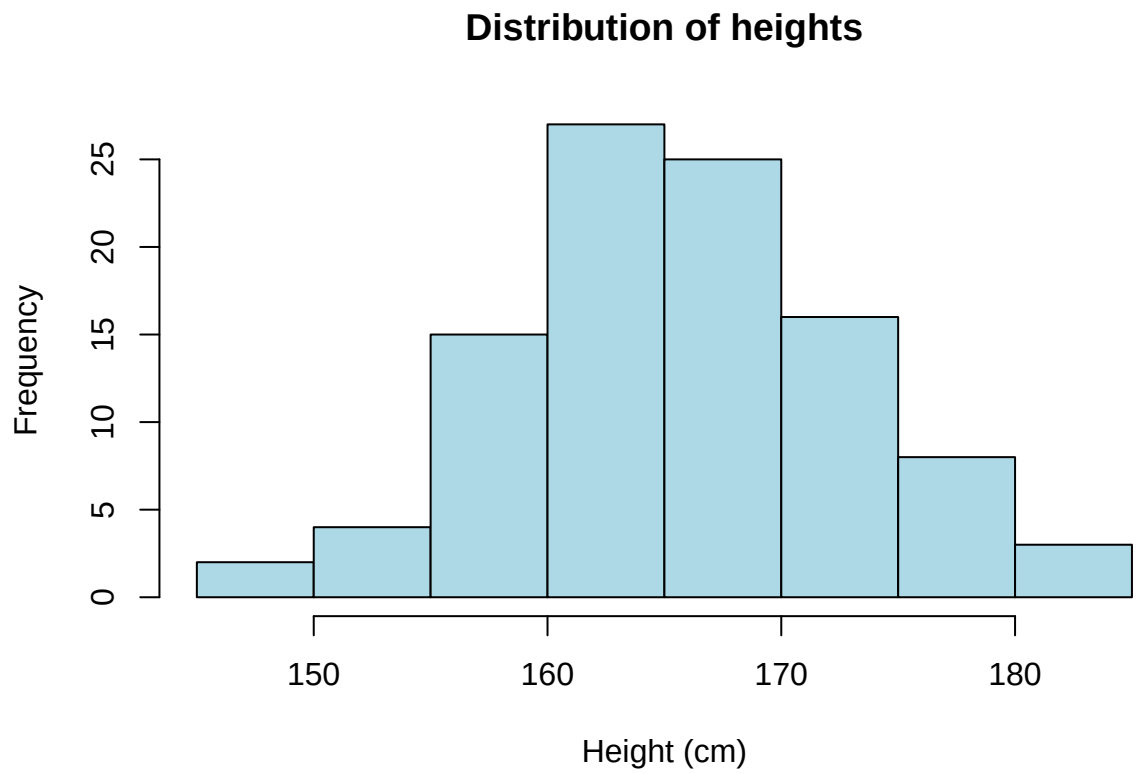



```
rm(list = ls())
# nesting ls inside rm

set.seed(123)  # reproducibility
```

1. What is variability?

Simulated heights of 100 individuals (cm)

```
heights <- rnorm (n = 100, mean = 165, sd = 8)

hist(heights,
     col = "lightblue",
     main = "Distribution of heights",
     xlab = "Height (cm)")
```



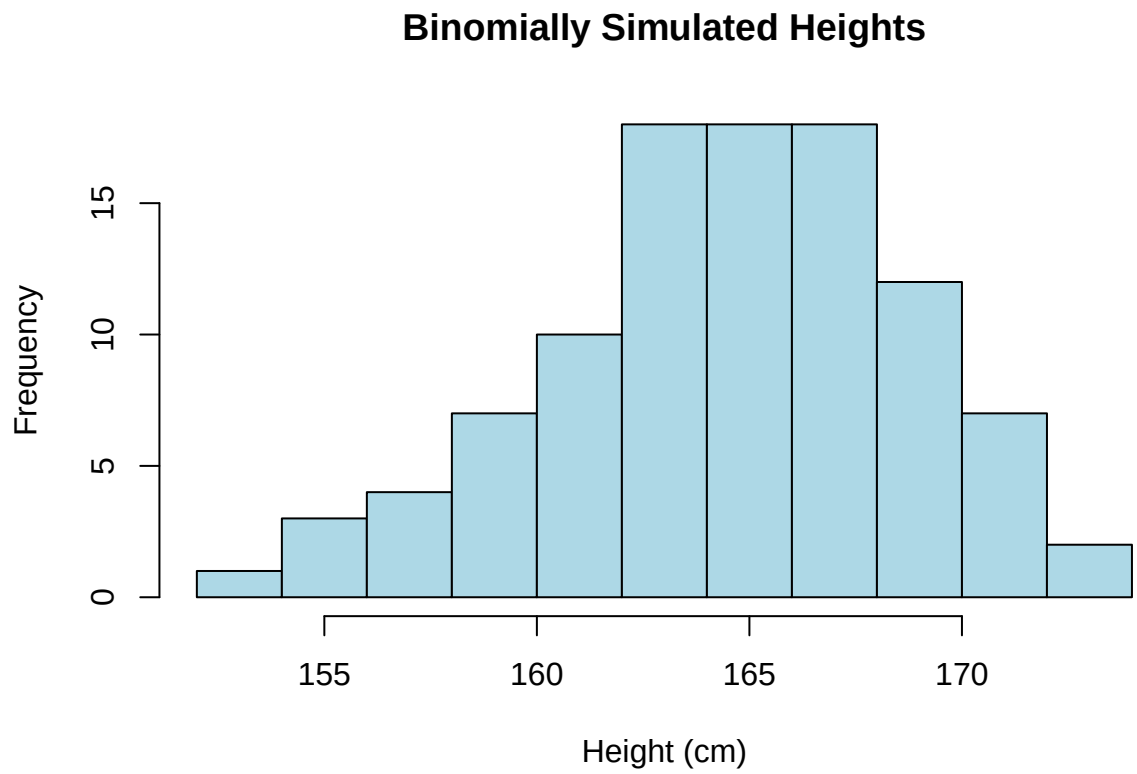

```r
mean(heights)
```

```
## [1] 165.7232
```

```r
sd(heights)
```

```
## [1] 7.302527
```

```r
heights <- rbinom(n = 100, size = 200, prob = 165/200)

hist(heights,
     col = "lightblue",
     main = "Binomially Simulated Heights",
     xlab = "Height (cm)")
```




```r
mean(heights)
```

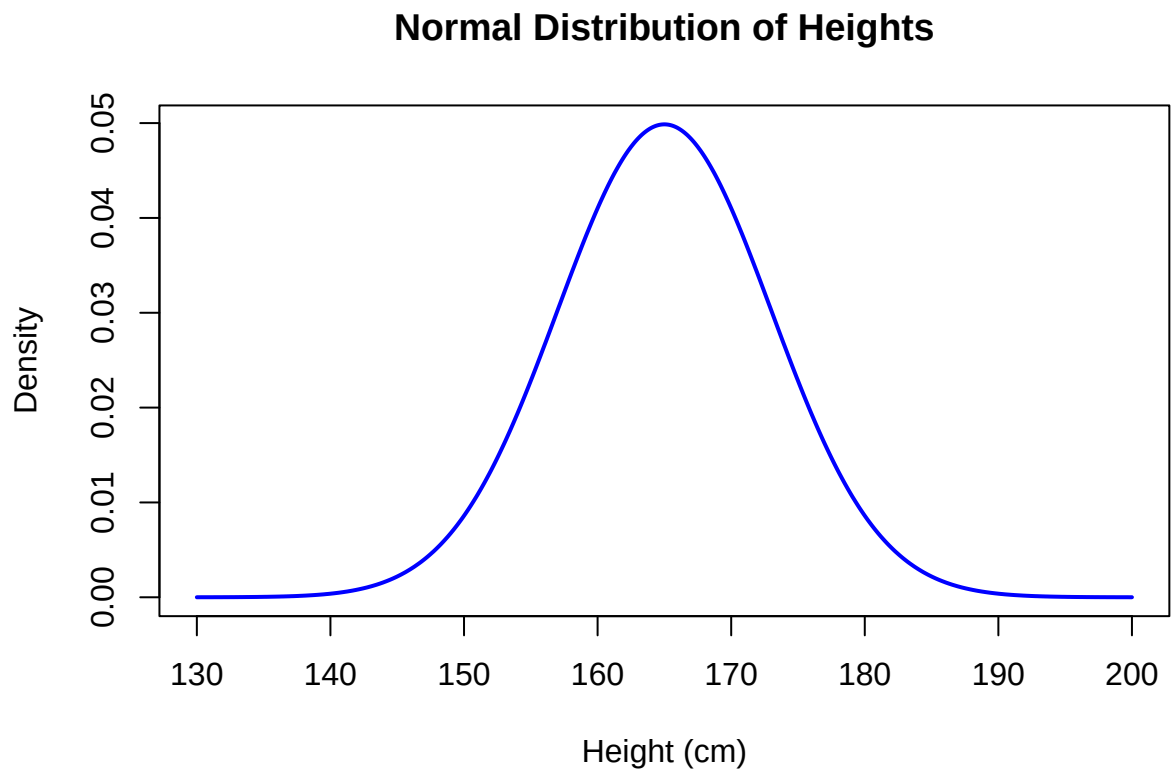
```
## [1] 164.94
```

```r
sd(heights)
```

```
## [1] 4.235063
```

2. Normal Distribution

```r
 x <- seq(130, 200, by = 0.1)
y <- dnorm(x, mean = 165, sd = 8)

plot(x, y,
     type = "l",
     lwd = 2,
     col = "blue",
     main = "Normal Distribution of Heights",
     xlab = "Height (cm)",
     ylab = "Density")
```


Normal Distribution of Heights
Density
0.00
0.01
0.02
0.03
0.04
0.05
130
140
150
160
170
180
190
200
Height (cm)